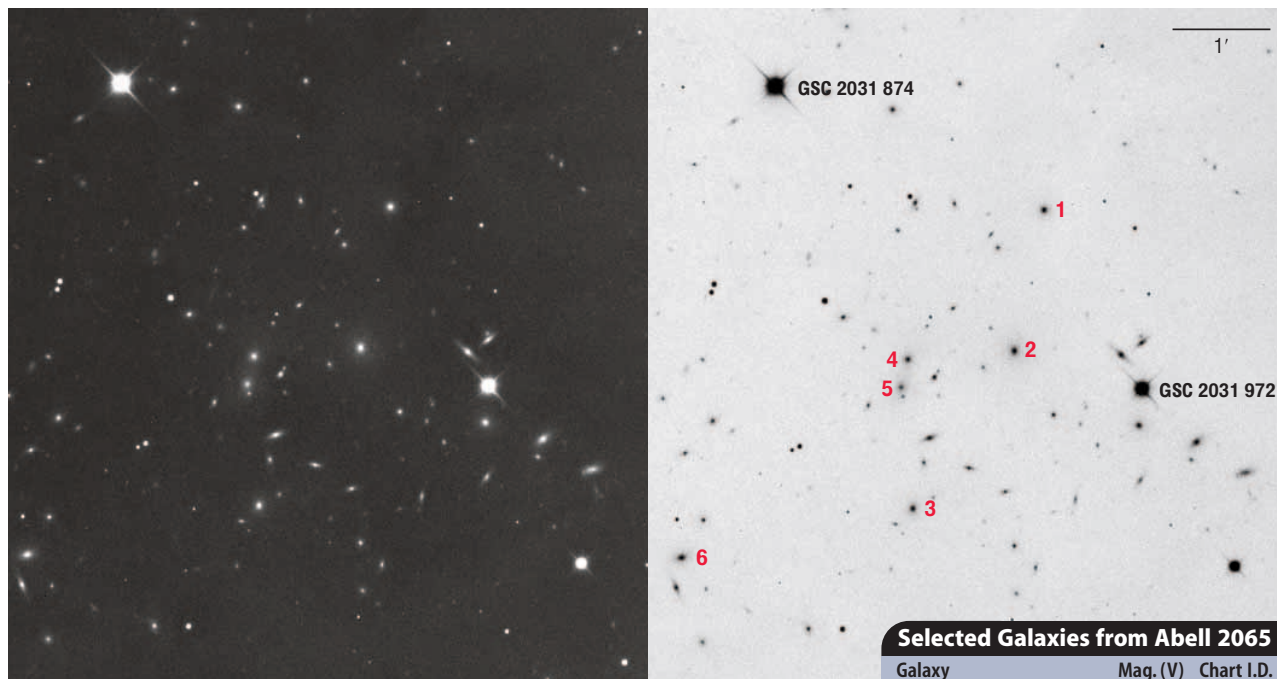




■ deep-sky notebook | By Steve Gottlieb

On the Edge — The Corona Borealis Galaxy Cluster

These are exciting times for deep-sky observers. Many objects rarely observed just a few decades ago have become familiar telescopic sights at today's star parties. For example, the Veil, Eagle, Rosette, and Horsehead nebulae were all once considered extremely difficult targets. But modern eye-

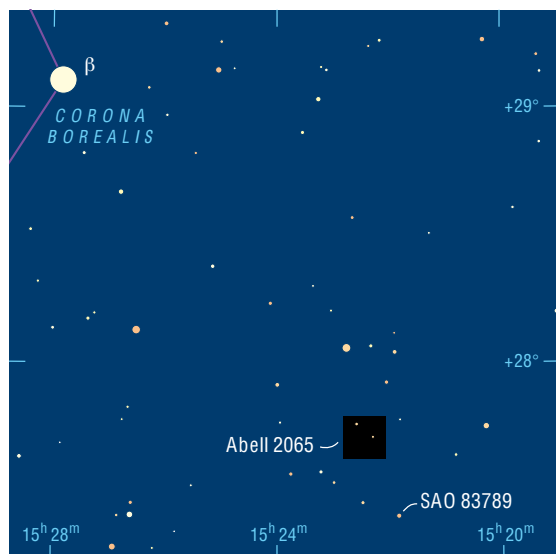
pieces, narrow-bandpass nebula filters, large transportable optics, and specialized observing charts have changed the kinds of objects regarded as observing challenges. Consider the Palomar globulars and Abell planetaries discovered in the mid-1950s on the *Palomar Observatory Sky Survey* plates. They were initially deemed beyond visual observation, yet during the 1970s determined amateurs were able to knock off a high percentage of these difficult objects.

So where is the cutting edge of deep-sky observing today? Some might say it lies in Corona Borealis. A well-known deep-sky entity that has resisted large-scale amateur assault is Abell 2065 — the Coro-

na Borealis galaxy cluster. In a list of deep-sky-challenge objects published in *The Observer's Handbook* of the Royal Astronomical Society of Canada, Abell 2065 is described as "perhaps the most difficult object for amateur telescopes."

I first became interested in observing this cluster nearly 20 years ago after I saw a deep photograph taken with the Mount Wilson 100-inch telescope. The field was peppered with dozens of tiny spindles and ovals floating among a pair of field stars. The diminutive size and extreme faintness of these galaxies result from their staggering distance — over 1 billion light-years away. Unfortunately, this remote sea of galaxies was well beyond the reach of my modest 8-inch Schmidt-Cassegrain telescope.

During the mid-80s I moved up to a 17.5-inch Dobsonian but was discouraged when I read in the *Webb Society Deep-Sky*



Left: The galaxy cluster Abell 2065 is located a short star-hop away from Beta (β) Coronae Borealis. **Above:** The central portion of this galaxy cluster is shown in this 30-minute exposure in blue light that was recorded with the Lick Observatory 120-inch telescope on June 26, 1960. Use the finder chart (above, right) and the accompanying table to locate and identify cluster members.

Selected Galaxies from Abell 2065

Galaxy	Mag. (V)	Chart I.D.
PGC 54846	15.3	—
MCG +5-36-19 (PGC 54875)	15.9	1
MCG +5-36-20 (PGC 54876)	15.5	2
MCG +5-36-21 (PGC 54084)	16.1	3
MCG +5-36-22	15.8	4
PGC 54883	15.6	5
MCG +5-36-23 (PGC 54891)	15.8	6

Observer's Handbook (Vol. 5: *Clusters of Galaxies*) that Abell 2065 was “undetectable in 16-inch reflectors even under good sky conditions.” As if that weren’t discouraging enough, the notation went on to mention that even in a 36-inch telescope the galaxy cluster was “very difficult.” Although Abell 2065 wasn’t mentioned in the deep-sky articles of the day, both the *SkyAtlas 2000.0* and *Uranometria 2000.0* star atlases plotted the cluster as if to suggest it was a reasonable target for amateur telescopes. I still dreamed about snagging the Corona Borealis cluster.

An opportunity to see the galaxy cluster finally arrived when I took a position as a teaching assistant for the San Francisco State University (SFSU) practical observational astronomy class. The week-long laboratory course was held at SFSU’s Sierra Nevada Field Campus at an elevation of 7,200 feet. After a few nights under transparent skies, I knew this site was up to the challenge of Abell 2065; I simply needed to arm myself with the necessary background information.

Abell 2065 consists of several hundred E- and S0-type galaxies arranged in a roughly spherical cloud with a condensed core region. A seminal 1956 *Astronomical Journal* article on galaxy redshifts by Milton Humason, Nicholas Mayall, and Allan Sandage listed eight members of Abell 2065 with redshifts of about 21,500 kilometers per second. Research on eight additional members was published in 1977, and all 16 galaxies were given *Principal Galaxy Catalogue* (PGC) designations. For magnitude information, I located a 1979 *Monthly Notices of the Royal Astronomical Society* paper that listed magnitudes for 156 cluster members along with an identification chart. More useful information was found in a 1988 *Astronomical Journal* article by Marc Postman, Margaret Geller, and John Huchra entitled “The Dynamics of the Corona Borealis Supercluster.”

With this information and a photograph of the cluster in hand, I patiently waited for a class session to coincide with a night of exceptional transparency. Finally such a night arrived and as soon as class ended, I immediately pointed my Dobsonian telescope toward Corona Borealis. The cluster’s core is situated less than 3° west-northwest of 2nd-magnitude Alpha (α) Coronae Borealis (Alphekka). Star-hopping to the field was easy, and I quickly located the 11th- and 12th-magnitude stars superimposed on

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the galaxy cluster's core.

The view at 144× was initially disappointing. No galaxies were immediately visible, but I was certain I had glimpsed some extremely faint nebulous knots near the 12th-magnitude star. I boosted the magnification up to 280×, checked the 100-inch photo, and prepared to get down to serious hunting.

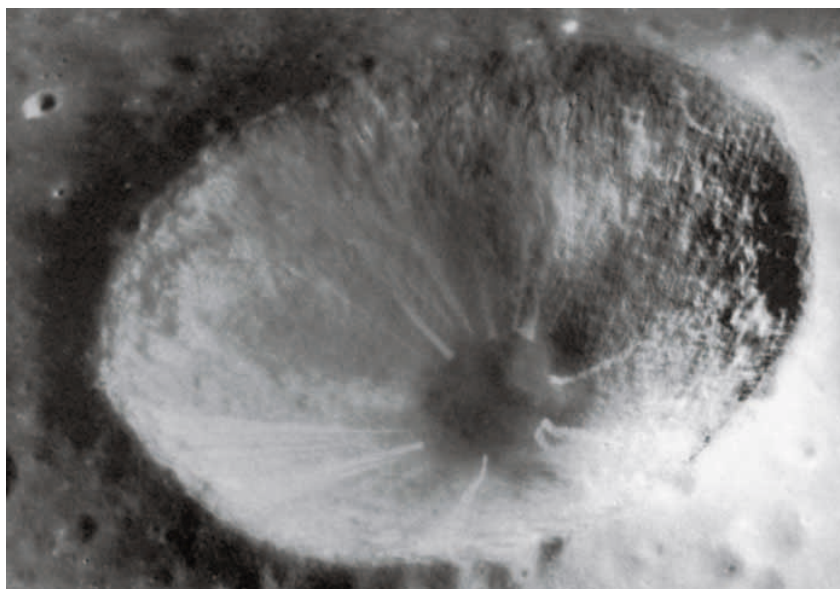
The easiest galaxy to find was MCG +5-36-20, located 1.5' east-northeast of the 12th-magnitude star. With averted vision I could steadily hold a 15-arcsecond hazy spot. Just 1.5' to the north lay MCG +5-36-19, a very difficult object. It could be intermittently glimpsed and appeared barely nonstellar.

Returning to the area immediately surrounding MCG +5-36-20, I noticed a slightly larger glow that was the combined light of MCG +5-36-22 and PGC 54883. In a moment of steady seeing, the haze split in two and the pair was cleanly resolved. With averted vision, the difficult galaxy MCG +5-36-21 popped into view 1.5' to the south of the duo. Last, I recorded MCG +5-36-23, a 16th-magnitude glow located 3' southeast of the close pair. All of these galaxies were contained in a 5' circle of sky.

After a solid hour and a half of visual exertion, I had managed to positively identify six members down to 16th magnitude. All were barely nonstellar and too faint to reveal any detail; the thrill was in seeing these distant objects at all! Surprisingly, PGC 54846, the brightest member listed in the *Dynamics* paper, is located nearly 20' south-southwest of the central core of the cluster. The galaxy is positioned just 1.5' northeast of the 9th-magnitude star SAO 83789 and is visible with direct vision as a 20" knot of light.

The key to successfully tackling Abell 2065 is transparent skies and at least a 12.5-inch telescope (Ronald Morales described his successful observations with a telescope of this size in the May 1990 *Sky & Telescope*, page 563) and lots of patience. Because these tiny galaxies are crammed into a field only a few arcminutes across, I recommend 200× to 300× to put some space between the individual members. Linger several minutes on each faint glow — savor the satisfaction of detecting photons that started their journey over 1 billion years ago.

A regular Deep-Sky Notebook contributor, Steve Gottlieb chases faint fuzzies from dark-sky sites in California.



In the main sequence of cratering morphology, Lalande A is categorized as a "simple" crater. Humble craters such as this have become standard tools for investigating a planet's history. For example, crater counts can be used to estimate the age of a planet's surface. Apollo 16 image courtesy Brown/NASA Northeast Regional Planetary Data Center.

■ lunar notebook | By Charles A. Wood

Lalande A — A Simple Crater

Impact craters are as common in the solar system as sand grains in Florida. As spacecraft have explored farther into space, impact craters have been discovered on virtually every world with a solid surface. There must be billions of impact craters of microscopic size and larger in the solar system. They result from cosmic collisions repeated ad infinitum, or as some might say, ad nauseam.

Most early selenographers considered

the variety of crater morphologies to represent fundamentally different types of structures. Not really understanding how craters formed, early Moon mappers demonstrated their scientific mettle by classifying craters into types with a plethora of different names: walled plains, mountain rings, ringed plains, crater plains, craters, craterlets, crater cones, crater pits, obscure rings, and depressions. While these various designations generally reflect differences in crater details (often due to later modification), they in fact obscure the basic underlying progression in crater morphology, which is simply a function of increasing diameter. Each lunar impact crater is unique, just as each human is, but there are only a few standard types of craters. From smallest to largest, impact craters have either simple, complex, or basin morphologies. This progression has been termed the main sequence of impact craters by lunar geologists in the U.S. Geological Survey.

Most lunar craters are of the simple variety in main-sequence terminology. A good example is Lalande A, a 13-kilometer-wide, 2.6-km-deep, fresh-looking im-



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